HealthConnect Rwanda

Healthcare Analytics and AI Transformation Programme

Project Management Final Report

AnalystLab Africa Experience Lab Internship Programme

Week 6 Assignment | Track: Project Management

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Submitted on: 25 Oct 2026

Version: 1.0 - Prepared for submission

This report is a planning package submitted in response to the official AnalystLab Africa Week 6 assignment. It contains proposed planning artefacts. No implementation, deployment, approval or benefit realisation is claimed. No patient data is included.

Document Control

Item	Detail
Document Title	HealthConnect Rwanda Healthcare Analytics and AI Transformation Programme - Project Management Final Report
Version	1.0 - Prepared for submission
Status	Prepared for submission - pending submission
Author	Lamiaa Samraoui, Project Management Intern
Reviewer	AnalystLab Africa
Classification	Academic submission - no patient data included
Reference Assignment	AnalystLab Africa Week 6 Assignment,
Planning Horizon A	Internship assignment - 7 calendar days (Timeline A)
Planning Horizon B	Proposed implementation - 24 weeks (Timeline B)

Version History

Version	Date	Author	Change summary
0.1	Day 1	PM Intern	Initial requirements extraction from PDF
0.2	Day 2	PM Intern	Business Case, Charter, Stakeholder Register
0.3	Day 3	PM Intern	Scope, WBS, Schedule integrated
0.4	Day 4	PM Intern	Resource Plan, Budget, Risk Register
0.5	Day 5	PM Intern	Comms, Quality, Backlog, Sprint Plan
0.6	Day 6	PM Intern	Change Mgmt, Dashboard, Closure
0.9	Day 7 AM	PM Intern	Executive Summary + Presentation
1.0	Day 7 PM	PM Intern	QA + two-timeline + cross-track register; prepared for submission

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1 Executive Summary

1.1 Overview

HealthConnect Rwanda has approved a strategic initiative to implement an integrated Healthcare Analytics and Artificial Intelligence platform. The platform would integrate data from seven source systems: Electronic Medical Records (EMR), Hospital Information Systems (HIS), Laboratory Information Systems (LIS), Pharmacy Management Systems, Billing Systems, Insurance Systems and Patient Feedback Platforms. The official assignment describes a relational database of 27 core datasets.

This report, the Control Workbook and the Executive Presentation form the 19-deliverable Project Management package (D-01 to D-19). Two timelines are used throughout. Timeline A is the 7-day internship assignment (INT-01 to INT-26). Timeline B is the proposed 24-week HealthConnect implementation. The two are never conflated.

1.2 Business Problem

The following eight problem statements are to be validated in the proposed Discovery phase; they are not measured findings.

1. Data is held in seven disconnected systems with no single integrated view.
2. Executives lack one consistent set of KPI definitions.
3. Management reports are assembled manually and slowly.
4. Patient, facility and department identifiers may not be consistent across systems.
5. Revenue, billing and insurance claim visibility is limited.
6. Bed, staff and equipment capacity decisions lack consolidated evidence.
7. Patient feedback is not linked to operational and clinical performance.
8. There is no governed route to safely introduce predictive models and generative AI.

In short: these are problem statements to be validated in the proposed Discovery phase; they are not measured findings.

1.3 Recommended Option

Option B - a phased 24-week integrated platform - is recommended over Option A (do minimum), Option C (big-bang) and Option D (vendor-in-a-box). Option B balances value, risk and governance by delivering in stages with stage gates (Section 2.3).

1.4 Delivery Structure

Delivery is organised into 12 WBS phases (P1–P12) over 24 weeks (Timeline B), with 11 milestones (MS-01 to MS-11) and four two-week sprints in Weeks 9–16. A hybrid waterfall-Agile model is used: waterfall for initiation, data collection and cleaning, testing and deployment; Agile Scrum for the build work in W9–W16.

1.5 Governance

A Steering Committee meets bi-weekly and is chaired by the Executive Sponsor (role TBC). The PMO meets weekly. Four Stage Gates control progress: Charter Approval (MS-01), Requirements Baseline (MS-02), Deployment Go/No-Go (MS-09) and Project Closure (MS-11).

1.6 Key Risks

The top five risks, scored as Probability × Impact, are: R-01 Source data quality (score 20, Critical); R-14 Integration complexity (16, Critical); R-05 Regulatory or data-protection non-compliance (15, Critical); R-06 Cybersecurity threat (15, Critical); and R-07 Clinical user change resistance (12, High). All 25 risks are in Appendix C.

1.7 Cross-Track Coordination

The Cross-Track Dependency Register (DEP-01 to DEP-18, Appendix B) records inputs expected from the Data Analytics, Data Science, ML Engineering and GenAI tracks. All 18 are PROVISIONAL and have the status Awaiting input; none has an evidence item attached.

1.8 Responsible AI

Human oversight is mandatory for any AI output affecting clinical or financial decisions (AS-09). A Data Protection Impact Assessment is required before deployment (REQ-RAI-009). The plan also requires SHAP/LIME explainability, model monitoring, bias and fairness testing, and hallucination red-team testing of GenAI assistants.

1.9 Success Criteria

Success is defined by SC-01 to SC-08 (Section 3.11). Benefits BEN-01 to BEN-09 would be tracked for 12 months after closure. No benefit is claimed as realised.

1.10 Decisions Required

1. Approve Option B (TBC).

2. Confirm the 24-week duration (TBC).

3. Confirm read-only integration for v1 (TBC).

4. Appoint the Executive Sponsor, Clinical Lead and Data Lead (TBC).

1.11 Integrity Statement

This report distinguishes four kinds of content: official assignment facts; proposed planning assumptions (labelled AS-xx); illustrative examples (labelled ILLUSTRATIVE); and provisional cross-track findings (Awaiting input). No approvals, named stakeholders, contracts, measured savings, model accuracy, completed milestones, deployments or actual expenditure are claimed.

2 Project Context and Business Case

2.1 Project Context

The official assignment places the Project Manager in charge of planning, executing, monitoring and delivering an enterprise healthcare analytics initiative that integrates data, improves efficiency and care, supports executive decisions and introduces AI. The project must be delivered on time, within budget and scope, while managing risk, communication, quality and change.

2.2 Business Problem

The eight problem statements in Section 1.2 apply. Proposed validation sources in Discovery are: stakeholder interviews (SH-01 to SH-20); profiling of the 27 core datasets (REQ-F-002); a sample of manual report preparation effort; finance reconciliation of revenue and claims; and a review of existing KPI definitions. Until validated they remain problem statements.

2.3 Options Analysis

Option	Description	Verdict
A	Do minimum: keep seven systems separate; ad-hoc manual reporting	Not recommended: leaves fragmentation and manual effort unresolved

B	Phased 24-week integrated platform: integrate, dashboards, ML, GenAI, with stage gates	RECOMMENDED: balances value, risk and governance
C	Big-bang: deliver all capabilities in one release	Not recommended: high delivery and adoption risk (R-14, R-15)
D	Vendor-in-a-box: buy a single packaged platform	Not recommended: lock-in, limited fit to seven local systems, residency uncertainty

2.4 Expected Benefits

Baselines and targets are TBC and would be measured in Discovery. No benefit is claimed as realised.

ID	Benefit	KPI	Baseline	Target
BEN-01	Faster executive reporting	Time to produce monthly executive report	TBC - measure in Discovery	TBC
BEN-02	Single integrated view of seven source systems	Systems integrated (of 7)	0 of 7	7 of 7 (PROPOSED)
BEN-03	Improved data quality	Data quality coverage % (KPI-008)	TBC - measure in Discovery	TBC
BEN-04	Faster financial reporting	Days to close monthly revenue report	TBC - measure in Discovery	TBC

BEN-05	Better insurance claim visibility	Claims approval rate visible by insurer	TBC	TBC
BEN-06	Better bed and resource visibility	Bed utilisation reported by facility	TBC	TBC
BEN-07	Patient satisfaction monitoring	Satisfaction reported by facility and department	TBC	TBC
BEN-08	Reduced manual reporting effort	Staff hours on manual reporting	TBC - measure in Discovery	TBC
BEN-09	Evidence-based decisions and adoption	Active dashboard users %	TBC	TBC

2.5 Cost Framework

All amounts are PLANNING PLACEHOLDER - NOT APPROVED, in RWF, and are not supplier quotes. Cost categories:

· Personnel (11 roles)

· Infrastructure and cloud services

· Software licences

· Data storage

· AI development

· Testing and security

· Deployment

· Training

· Contingency (10% of direct cost, AS-08)

· Management reserve (5% of direct cost, AS-08)

2.6 ROI Methodology

ROI would be calculated once Finance validates costs and benefits. Proposed formulas (discount rate r = 12%, PROPOSED, AS-12):

· NPV = sum over periods t of (Benefit_t − Cost_t) / (1 + r)^t

· Payback period = time at which cumulative net benefit becomes positive

· Benefit-Cost Ratio (BCR) = PV of benefits / PV of costs

· ROI % = (Total benefits − Total costs) / Total costs × 100

Three scenarios are defined for sensitivity analysis. They are ILLUSTRATIVE and contain no figures: Conservative (benefits realised more slowly and below the base estimate), Base (benefits as estimated by Finance), Optimistic (benefits realised faster and above the base estimate).

2.7 Strategic Alignment

Pillar	Contribution
Improve healthcare service delivery	Single view of clinical and operational data for timely decisions
Optimise hospital operations	Bed, staff and equipment visibility across facilities
Enhance patient outcomes	Clinical analytics and governed predictive models with human oversight
Support financial sustainability	Revenue, billing and claims reporting by facility, department and insurer
Accelerate digital transformation	Integrated platform with Responsible AI governance

2.8 Major Risks and Constraints

Top risks: R-01 (20), R-14 (16), R-05 (15), R-06 (15) and R-07 (12). Constraints: (1) the 24-week duration is proposed, not approved; (2) all budget figures are placeholders; (3) integration is read-only in v1 (AS-05); (4) data must reside in Rwanda (AS-10) and comply with applicable data-protection requirements.

2.9 Recommendation

Approve Option B, subject to Finance validation of the placeholder budget and appointment of the Sponsor, Clinical Lead and Data Lead before Gate 1.

3 Project Charter

3.1 Purpose

This Charter proposes the authority, objectives, scope, governance and success criteria for the HealthConnect Rwanda Healthcare Analytics and AI Transformation Programme. It is PROPOSED and unsigned.

3.2 Business Need

See Section 2.2. The need is to integrate seven source systems and enable analytics and AI safely for executives, clinicians and finance users.

3.3 Objectives

ID	Objective	Measure	Timeline	BO link
OBJ-01	Integrate the seven source systems into one read-only warehouse	7 of 7 systems integrated; counts reconciled	By Week 10	BO-01
OBJ-02	Deliver five role-based dashboards	5 dashboards pass UAT	By Week 14	BO-02, BO-04, BO-05

OBJ-03	Evaluate three predictive models with explainability	Metrics vs agreed thresholds; SHAP/LIME report	By Week 16	BO-03, BO-08
OBJ-04	Pilot governed GenAI assistants with human oversight	Evaluation report; red-team passed	By Week 18	BO-07
OBJ-05	Complete Responsible AI and DPIA before deployment	DPIA signed; RAI pack complete	By Week 19	BO-03, BO-07
OBJ-06	Deploy and train users	Go/No-Go approved; training completion at target	By Week 23	BO-05, BO-06
OBJ-07	Close the project with benefits tracking set up	Closure criteria CLO-01 to CLO-10 met	By Week 24	BO-06

Business objectives: BO-01 Integrate enterprise healthcare datasets into a unified analytics platform; BO-02 Develop executive dashboards; BO-03 Improve patient care through analytics and AI; BO-04 Enhance financial reporting; BO-05 Improve operational efficiency; BO-06 Support data-driven decision-making; BO-07 Develop AI-powered healthcare assistants; BO-08 Build predictive healthcare models.

3.4 High-Level Requirements

· Functional (20): ingestion, cleaning, dashboards, predictive models, GenAI assistants, PMO dashboard, training

· Non-functional (10): performance, availability, accessibility, localisation, freshness

· Data (8): lineage, quality, identifiers, retention, residency

· Security (8): RBAC, MFA, encryption, audit, penetration testing

· Responsible AI (10): oversight, explainability, fairness, DPIA, governance

· Operational (4): hypercare, runbooks, support, handover

3.5 Scope Summary

In scope (15 items): read-only extraction from seven systems, profiling and cleaning of 27 datasets, integrated warehouse, batch pipelines, dashboards, three predictive models, GenAI pilots on governed sources, Responsible AI artefacts, testing, deployment, training and closure.

Out of scope (10 items): write-back to source systems, replacement of source systems, hardware procurement, autonomous AI clinical decisions, device certification, national data exchange, operations after hypercare, new data sources, SMS notifications and real-time streaming (the last two deferred to v2).

3.6 Major Deliverables

ID	Deliverable	Owner	Due (Timeline B)	Submission (Timeline A)
D-01	Business Case	Project Manager	Week 2 (relative)	Day 2
D-02	Project Charter	Project Manager	Week 2 (relative)	Day 2
D-03	Stakeholder Register	Business Analyst	Week 2 (relative)	Day 2
D-04	Scope Statement	Business Analyst	Week 4 (relative)	Day 3
D-05	Work Breakdown Structure	Project Manager	Week 4 (relative)	Day 3
D-06	Gantt Chart / Project Schedule	Project Manager	Week 4 (relative)	Day 3
D-07	Resource Plan	Project Manager	Week 4 (relative)	Day 4

D-08	Budget Estimate	Project Manager	Week 4 (relative)	Day 4
D-09	Risk Register	Project Manager	Week 4 (relative)	Day 4
D-10	Probability–Impact Matrix	Project Manager	Week 4 (relative)	Day 4
D-11	Communication Plan	Project Manager	Week 4 (relative)	Day 5
D-12	Quality Management Plan	QA/Test Engineer	Week 4 (relative)	Day 5
D-13	Product Backlog	Business Analyst	Week 4 (relative)	Day 5
D-14	Sprint Plan	Project Manager	Week 4 (relative)	Day 5
D-15	Change Management Plan	Change Lead	Week 4 (relative)	Day 6
D-16	Project Dashboard	Project Manager	Week 4 (relative)	Day 6
D-17	Project Closure Report	Project Manager	Week 24 (relative)	Day 6
D-18	Executive Summary (3–5 pages)	Project Manager	Week 2 (relative)	Day 7
D-19	Presentation Slides (10–15 slides)	Project Manager	Week 2 (relative)	Day 7

3.7 Proposed Milestones

ID	Milestone	Due	Gate
MS-01	Project Charter approved (Stage Gate 1 - Charter Approval)	Week 2	Gate 1
MS-02	Requirements and scope baselined (Stage Gate 2 - Requirements Baseline)	Week 4	Gate 2
MS-03	Data quality report accepted	Week 8	-
MS-04	Integrated data warehouse ready for analytics	Week 10	-
MS-05	First two dashboards (Executive, Financial) signed off	Week 12	-
MS-06	Model Review Gate - all ML models evaluated	Week 16	-
MS-07	GenAI evaluation complete	Week 18	-
MS-08	Responsible AI assessment and DPIA signed	Week 19	-
MS-09	Deployment Go/No-Go (Stage Gate 3)	Week 20	Gate 3
MS-10	User training complete	Week 23	-
MS-11	Project closure and sign-off (Stage Gate 4 - Project Closure)	Week 24	Gate 4

3.8 Budget Estimate Status

The budget is a set of planning placeholders in RWF (PLANNING PLACEHOLDER - NOT APPROVED). The Budget sheet computes a direct total from personnel person-weeks and leaves non-personnel quantities AWAITING INPUT. Contingency is 10% and management reserve 5% of direct cost (AS-08). No amount is approved or spent.

3.9 Assumptions and Constraints

ID	Assumption	Confidence
AS-01	Proposed implementation duration is 24 weeks (Timeline B)	Medium
AS-02	Four Stage Gates and 11 milestones structure delivery	Medium
AS-03	Four two-week sprints run in W9–W16	Medium
AS-04	Source-system owners grant read access	Low
AS-05	Integration in v1 is read-only (no write-back)	High
AS-06	Single hyperscaler cloud provider, to be confirmed	Low
AS-07	All planning figures are RWF placeholders	High
AS-08	Contingency 10% and management reserve 5% of direct cost	Medium
AS-09	Human oversight of AI is mandatory for clinical or financial outputs	High

AS-10	Data residency is in Rwanda	High
AS-11	27 core datasets as stated in the assignment	High
AS-12	NPV discount rate 12% (PROPOSED)	Low
AS-13	Batch data freshness of 24 hours or less; no streaming in v1	Medium
AS-14	Sponsor, Clinical Lead and Data Lead appointed before Gate 1	Low
AS-15	Weekly clinical review slot is available	Medium
AS-16	Key stakeholders available in W3–W4	Medium
AS-17	Localisation in English, French and Kinyarwanda	Medium
AS-18	The 12-phase WBS fits within 24 weeks	Medium
AS-19	Working week is 5 days	High
AS-20	Data Analytics and Data Science track inputs are PROVISIONAL	High
AS-21	ML Engineering track inputs are PROVISIONAL	High
AS-22	GenAI track inputs are PROVISIONAL	High

3.10 Roles

Role	Responsibility	Status
Executive Sponsor	Chairs SteerCo; approves gates and decisions	TBC
Project Manager	Plans, controls and reports	Intern role (this package)
Clinical Lead	Clinical safety review and oversight	TBC
Data Lead	Source data, quality and integration	TBC
Analytics Lead	KPI definitions and dashboards	TBC
ML Lead	Models and evaluation	TBC
GenAI Lead	Prompts, RAG and assistants	TBC
Legal/DPO	Privacy, DPIA, regulatory review	TBC
Finance Lead	Budget validation and benefits	TBC
IT/Security Lead	Infrastructure and security	TBC

3.11 Success Criteria

ID	Criterion
SC-01	All 7 source systems integrated read-only with counts reconciled
SC-02	Data quality report covers all 27 core datasets
SC-03	Five dashboards pass UAT at or above the agreed pass rate
SC-04	Three ML models evaluated with explainability reports
SC-05	GenAI pilots evaluated for hallucination and citation quality
SC-06	DPIA signed and Responsible AI pack complete before deployment
SC-07	Training completion at or above target (KPI-007)
SC-08	Closure criteria met and benefits tracking active for 12 months post-closure

3.12 Governance and Escalation

The Executive Sponsor chairs the Steering Committee (bi-weekly). The PMO meets weekly. Four Stage Gates apply: Charter Approval, Requirements Baseline, Deployment Go/No-Go and Project Closure. Escalation follows four levels (Section 9.3). Change requests follow Section 11.1.

3.13 Assignment vs Implementation Timeline

Aspect	Timeline A - internship assignment	Timeline B - proposed implementation

Purpose	Prepare the PM planning package	Proposed delivery of the platform
Duration	7 calendar days	24 weeks
Identifiers	INT-01 to INT-26 (Day 1 INT-01–04; Day 2 INT-05–08; Day 3 INT-09–12; Day 4 INT-13–17; Day 5 INT-18–21; Day 6 INT-22–24; Day 7 INT-25–26)	Week 1 to Week 24; MS-01 to MS-11
Dates	DD/MM/YYYY only if confirmed; none invented	Relative weeks only
Status	Package prepared for submission	PROPOSED - not started; nothing implemented

3.14 Approval and Sign-Off

Blank block. No approval is claimed.

Role	Name	Signature	Date
Executive Sponsor			
Project Manager			
Clinical Lead			
Data Lead			

4 Stakeholder Analysis

4.1 Power-Interest Map

Influence \ Interest	Lower interest (1–3)	Higher interest (4–5)
Higher influence (4–5)	Keep Satisfied: SH-02, SH-09	Manage Closely: SH-01, SH-03, SH-10, SH-16, SH-17, SH-20
Lower influence (1–3)	Monitor: SH-08, SH-15, SH-18, SH-19	Keep Informed: SH-04, SH-05, SH-06, SH-07, SH-11, SH-12, SH-13, SH-14

4.2 Stakeholder Register Summary

ID	Stakeholder	Influence	Interest	Expectations
SH-01	Executive Board	5	5	Sponsor; evidence-based decisions and visible value
SH-02	Ministry of Health	5	3	Regulatory compliance; national data alignment
SH-03	Hospital Directors	4	5	Better facility performance visibility

SH-04	Doctors	3	5	Trustworthy, explainable decision support
SH-05	Nurses	2	5	Low-burden tools; workload visibility
SH-06	Pharmacists	2	4	Accurate medication data
SH-07	Laboratory Scientists	2	4	Reliable test volume and turnaround views
SH-08	Patients	2	3	Privacy and better care
SH-09	Insurance Providers	4	3	Accurate claims data and timely reporting
SH-10	IT Department	4	5	Secure, supportable integration
SH-11	Data Analysts	2	5	Clear KPI definitions and clean data
SH-12	Data Scientists	2	5	Reliable labels and cleaned data
SH-13	Machine Learning Engineers	2	5	Clear targets and thresholds
SH-14	Generative AI Engineers	2	5	Governed sources and clear guardrails

SH-15	Vendors	3	3	Clear requirements; timely payment
SH-16	Finance Department	4	4	Validated budget and reliable financial reporting
SH-17	Legal and Data Protection Office	4	4	Compliant design and DPIA completion
SH-18	Radiographers	1	3	Minimal disruption
SH-19	Administrative Staff	2	3	Simpler reporting; accurate records
SH-20	Project Management Office	4	5	Timely, accurate project data

4.3 Engagement Risks

Risk	Stakeholders	Register ID
Clinical user resistance to new tools	SH-04, SH-05, SH-06, SH-07	R-07
Delayed approvals and executive disengagement	SH-01, SH-03	R-04, R-19
Source system access and integration delays	SH-10, SH-15	R-09, R-14
Regulatory and privacy concerns	SH-02, SH-08, SH-17	R-05, R-11

4.4 Review Cadence

The register is reviewed weekly by the PMO and formally at each Stage Gate (Week 2, Week 4, Week 20, Week 24, relative). Scores are PROPOSED and would be validated in Discovery.

5 Scope and Requirements

5.1 Scope Statement

Product scope: an integrated, read-only healthcare analytics platform built on the seven source systems, providing five role-based dashboards, three evaluated predictive models and governed GenAI assistant pilots, with Responsible AI controls. Project scope: the work to plan, build, test, deploy, train users on and close that platform within the proposed 24 weeks (Timeline B).

5.2 In-Scope

ID	In-scope item
SC-IN-01	Read-only extraction from the seven source systems
SC-IN-02	Profiling and cleaning of the 27 core datasets
SC-IN-03	Integrated warehouse and data model
SC-IN-04	Batch ETL/ELT pipelines (freshness 24 hours or less)
SC-IN-05	Data quality report and foreign-key validation
SC-IN-06	Executive dashboard

SC-IN-07	Financial dashboard
SC-IN-08	Clinical dashboard
SC-IN-09	Operations and patient experience dashboards
SC-IN-10	Predictive models (three) with explainability
SC-IN-11	GenAI assistants and RAG pilots on governed sources
SC-IN-12	Responsible AI, DPIA and governance artefacts
SC-IN-13	Testing, UAT and security testing
SC-IN-14	Deployment and hypercare
SC-IN-15	User training and project closure

5.3 Out-of-Scope

ID	Out-of-scope item
SC-OUT-01	Write-back to source systems
SC-OUT-02	Replacement or upgrade of source systems

SC-OUT-03	Procurement of clinical hardware and end-user devices
SC-OUT-04	Autonomous AI clinical decision-making
SC-OUT-05	Medical-device regulatory certification of models
SC-OUT-06	National or MoH data-exchange integration
SC-OUT-07	Ongoing operations after hypercare
SC-OUT-08	New data sources beyond the seven systems
SC-OUT-09	SMS discharge notifications (deferred to v2 via CHG-01)
SC-OUT-10	Real-time streaming ingestion (deferred to v2)

5.4 Requirements Catalogue

Total = 60 requirements: 20 Functional, 10 Non-Functional, 8 Data, 8 Security, 10 Responsible AI and 4 Operational. Targets marked PROPOSED are to be validated.

Functional Requirements (20)

ID	Requirement	Priority	Owner
REQ-F-001	Ingest read-only extracts from the seven source systems	Must	Data Engineer

REQ-F-002	Profile all 27 core datasets	Must	Data Engineer
REQ-F-003	Standardise categories and resolve duplicate records	Must	Data Engineer
REQ-F-004	Verify foreign-key integrity and issue a data quality report	Must	Data Engineer
REQ-F-005	Provide an integrated warehouse data model	Must	Data Engineer
REQ-F-006	Executive dashboard with KPI cards for patients, appointments, revenue, facilities, doctors and satisfaction	Must	Dashboard Developer
REQ-F-007	Financial dashboard: revenue trend, by facility and department, claims, payment methods	Must	Dashboard Developer
REQ-F-008	Clinical dashboard: diagnoses, laboratory tests, prescriptions, admissions, referrals	Must	Dashboard Developer
REQ-F-009	Operations dashboard: staff, beds, equipment, facility and department comparison	Must	Dashboard Developer
REQ-F-010	Patient experience dashboard: ratings, feedback, hospital and department ranking	Should	Dashboard Developer
REQ-F-011	Slicers, drill-through and role-based views on all dashboards	Should	Dashboard Developer

REQ-F-012	Hospital admission prediction model	Must	ML Engineer
REQ-F-013	Patient satisfaction prediction model	Should	ML Engineer
REQ-F-014	Insurance claim approval prediction model	Should	ML Engineer
REQ-F-015	Prompt library of at least 20 reviewed prompts	Must	GenAI Engineer
REQ-F-016	RAG assistant answering executive and operational questions with citations from governed sources	Must	GenAI Engineer
REQ-F-017	AI agent workflow supporting administrators with reporting	Could	GenAI Engineer
REQ-F-018	AI-assisted monthly operational report draft with human review	Could	GenAI Engineer
REQ-F-019	PMO project dashboard tracking schedule, budget, risk, issues, milestones, resources, quality and stakeholders	Must	Project Manager
REQ-F-020	Training materials and sessions in English, French and Kinyarwanda	Must	Change Lead

Non-Functional Requirements (10)

ID	Requirement	Owner

REQ-NF-001	Dashboards load within 5 seconds for standard views (PROPOSED target)	Dashboard Developer
REQ-NF-002	Platform availability target of 99.5% in business hours (PROPOSED)	IT/Security Lead
REQ-NF-003	Scales to all participating facilities without redesign	Data Engineer
REQ-NF-004	Usability validated with clinical and executive users in UAT	Business Analyst
REQ-NF-005	Accessibility: readable contrast and keyboard navigation	Dashboard Developer
REQ-NF-006	Localisation in English, French and Kinyarwanda (AS-17)	Change Lead
REQ-NF-007	Documentation maintained and version controlled	Project Manager
REQ-NF-008	Backup and recovery tested before go-live	IT/Security Lead
REQ-NF-009	Batch data freshness of 24 hours or less (AS-13)	Data Engineer
REQ-NF-010	Open, documented data formats and interfaces	Data Engineer

Data Requirements (8)

ID	Requirement	Owner

REQ-DR-001	Record data lineage from source to dashboard	Data Engineer
REQ-DR-002	Define data quality thresholds per dataset	Data Engineer
REQ-DR-003	Maintain consistent patient, facility and department identifiers	Data Engineer
REQ-DR-004	Define retention rules for raw and integrated data	Legal/DPO
REQ-DR-005	Host data within Rwanda (AS-10)	IT/Security Lead
REQ-DR-006	Pseudonymise patient identifiers in analytics and AI layers	Data Engineer
REQ-DR-007	Publish a data dictionary for the integrated model	Data Engineer
REQ-DR-008	Reconcile record counts between source and warehouse each load	Data Engineer

Security Requirements (8)

ID	Requirement	Owner
REQ-SEC-001	Role-based access control for all platform components	IT/Security Lead
REQ-SEC-002	Multi-factor authentication for privileged users	IT/Security Lead

REQ-SEC-003	Encryption of data at rest and in transit	IT/Security Lead
REQ-SEC-004	Audit logging of data and dashboard access	IT/Security Lead
REQ-SEC-005	Penetration test passed before Go/No-Go	QA/Test Engineer
REQ-SEC-006	Secrets and key management	IT/Security Lead
REQ-SEC-007	Security incident response procedure	IT/Security Lead
REQ-SEC-008	Security review of vendors and third-party services	IT/Security Lead

Responsible AI Requirements (10)

ID	Requirement	Owner
REQ-RAI-001	Human oversight of AI outputs affecting clinical or financial decisions (AS-09)	Clinical Lead
REQ-RAI-002	Explainability reports (SHAP/LIME) for every model	ML Engineer
REQ-RAI-003	Bias and fairness testing across patient groups	ML Engineer

REQ-RAI-004	Hallucination red-team testing of GenAI assistants	GenAI Engineer
REQ-RAI-005	Model monitoring and drift detection with retraining triggers	ML Engineer
REQ-RAI-006	Label all AI-generated outputs as AI-generated	GenAI Engineer
REQ-RAI-007	No autonomous clinical decisions by AI	Clinical Lead
REQ-RAI-008	Model documentation (purpose, data, limits, metrics)	ML Engineer
REQ-RAI-009	Data Protection Impact Assessment completed before deployment	Legal/DPO
REQ-RAI-010	AI Governance Committee established with review authority	Legal/DPO

Operational Requirements (4)

ID	Requirement	Owner
REQ-OP-001	Hypercare support after go-live	IT/Security Lead
REQ-OP-002	Operational runbooks for pipelines and dashboards	Data Engineer
REQ-OP-003	Support model and service levels agreed with operations	IT/Security Lead

REQ-OP-004	Handover and knowledge transfer to operations	Project Manager

5.5 Requirements Traceability Matrix

BO	REQ	Scope	WBS	Deliverable	AC	Evidence	Owner
BO-01	REQ-F-001	SC-IN-01	WP-3.4	D-05	AC-01	EVD-05	Data Engineer
BO-02	REQ-F-006	SC-IN-06	WP-6.3	D-13	AC-05	EVD-13	Dashboard Developer
BO-08	REQ-F-012	SC-IN-10	WP-7.3	D-13	AC-06	EVD-13	ML Engineer
BO-06	REQ-SEC-005	SC-IN-13	WP-9.3	D-09	AC-09	EVD-09	QA/Test Engineer
BO-03	REQ-RAI-009	SC-IN-12	WP-8.7	D-09	AC-08	EVD-09	Legal/DPO

The full 60-row matrix is in workbook sheet Requirements_Traceability. All statuses are PROPOSED.

5.6 Acceptance Criteria Key

AC	Acceptance criterion	Evidence

AC-01	Read-only extraction from seven systems completed; record counts reconciled	EVD-05
AC-02	27 core datasets profiled; data quality report issued	EVD-12
AC-03	Foreign-key integrity verified; duplicates resolved	EVD-12
AC-04	Integrated warehouse accepted; reconciles to profiled counts	EVD-06
AC-05	Dashboards pass UAT checklist; KPI definitions signed	EVD-16
AC-06	Models meet agreed thresholds; explainability and monitoring documented	EVD-12
AC-07	GenAI evaluation shows hallucination and citation results; governed sources only	EVD-12
AC-08	Responsible AI pack complete; DPIA signed; oversight evidenced	EVD-09
AC-09	Security tests passed; no critical findings open	EVD-09
AC-10	Users trained; completion at or above target	EVD-15
AC-11	Deployment Go/No-Go approved; operational readiness evidenced	EVD-02
AC-12	Closure criteria CLO-01 to CLO-10 met; handover signed	EVD-17

5.7 Scope Baseline and Change Control

The scope baseline consists of this Scope Statement, the in/out lists, the requirements catalogue and the WBS. It is PROPOSED until Gate 2 (MS-02, Week 4). After baselining, any change follows the change control process in Section 11.1. CHG-01 shows how a deferral is recorded.

6 WBS and Schedule

6.1 WBS Overview

Phase	Work packages	Window (Timeline B)
P1 Project Initiation	5 (WP-1.1 to WP-1.5)	Weeks 1–2
P2 Requirements Gathering	15 (WP-2.1 to WP-2.15)	Weeks 2–4
P3 Data Collection	5 (WP-3.1 to WP-3.5)	Weeks 5–6
P4 Data Cleaning	5 (WP-4.1 to WP-4.5)	Weeks 7–8
P5 Data Integration	5 (WP-5.1 to WP-5.5)	Weeks 9–10
P6 Dashboard Development	9 (WP-6.1 to WP-6.9)	Weeks 11–14
P7 Machine Learning Development	8 (WP-7.1 to WP-7.8)	Weeks 13–16
P8 Generative AI Development	8 (WP-8.1 to WP-8.8)	Weeks 14–18

P9 Testing	6 (WP-9.1 to WP-9.6)	Weeks 17–20
P10 Deployment	4 (WP-10.1 to WP-10.4)	Weeks 21–22
P11 User Training	5 (WP-11.1 to WP-11.5)	Weeks 21–23
P12 Project Closure	5 (WP-12.1 to WP-12.5)	Weeks 23–24
Total	80 work packages	24 weeks

6.2 Schedule Overview

The schedule has 53 activities (ACT-1.1.1 to ACT-12.5.1) and 11 milestones across 24 relative weeks, with a sprint window in W9–W16. Four overlapping activities use Start-to-Start links: ACT-1.4.1, ACT-2.2.1, ACT-2.8.1 and ACT-7.2.1. Phases overlap deliberately (for example P6, P7 and P8 in Weeks 13–16). The Gantt chart is in workbook sheet Gantt_Chart; dates are relative weeks.

6.3 Indicative Critical Path

The indicative critical path runs through: ACT-1.1.1 → ACT-1.3.1 → ACT-1.4.1 → ACT-2.1.1 → ACT-2.2.1 → ACT-2.9.1 → ACT-3.2.1 → ACT-3.3.1 → ACT-4.1.1 → ACT-4.4.1 → ACT-5.1.1 → ACT-5.4.1 → ACT-6.3.1 → ACT-6.5.1 → ACT-6.8.1 → ACT-9.1.1 → ACT-9.2.1 → ACT-9.5.1 → ACT-10.3.1 → ACT-12.1.1 → ACT-12.5.1. It is indicative only and would be validated when durations are confirmed.

6.4 Dependencies That Make Sequence Non-Negotiable

· Source access and profiling (P3) must precede cleaning (P4) and integration (P5).

· Integrated data (MS-04) must precede dashboards, models and RAG indexing.

· Labels and features (DEP-04, DEP-06) must exist before model evaluation (DEP-05).

· DPIA and Responsible AI assessment (MS-08) must precede the Go/No-Go (MS-09).

· UAT and security testing must pass before deployment (P10).

6.5 Delay and Escalation Rules

· Variance of 0 weeks: Green. Slip of 1–3 weeks: Amber, Project Manager manages and PMO is informed.

· Slip of more than 3 weeks, or any slip on the critical path or a Stage Gate: Red, escalated to the Steering Committee.

· A slip above 5 working days on any approval is escalated to the Sponsor role.

7 Resource and Budget Plan

7.1 Human Resources

Role	FTE	Phases	Responsibilities
Project Manager	1.0	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P12	Plan, control, report, escalate
Business Analyst	1.0	P1, P2, P6, P9, P12	Requirements, UAT coordination
Data Engineer	2.0	P3, P4, P5, P9, P10	Extraction, cleaning, integration, pipelines
Dashboard Developer	2.0	P6, P9, P10	Dashboards and access controls
ML Engineer	1.0	P7, P9, P10	Models, explainability, monitoring

GenAI Engineer	1.0	P8, P9, P10	Prompts, RAG, assistants, evaluation
QA/Test Engineer	2.0	P6, P7, P8, P9, P10	Test strategy, SIT, UAT support
Change Lead	1.0	P1, P2, P6, P9, P10, P11, P12	Adoption, training, communications
Clinical Lead	0.5	P2, P6, P7, P8, P9, P11	Clinical review and safety oversight
IT/Security Lead	0.5	P2, P5, P9, P10	Infrastructure, security, access
Legal/DPO	0.3	P2, P7, P8, P9, P12	Privacy, DPIA, regulatory review

FTE are PROPOSED. Lead roles (Sponsor, Clinical, Data) are TBC.

7.2 Non-Human Resources

· Hardware: existing facility workstations; no hardware procurement is in scope (SC-OUT-03).

· Software: BI and dashboard tools, ETL/ELT tooling, AI and LLM platform (PLANNING PLACEHOLDER).

· Cloud services: single hyperscaler hosted in Rwanda (AS-06, AS-10, TBC).

· Environments: development, staging and production.

· Training: rooms and online sessions in English, French and Kinyarwanda.

7.3 Budget Workbook Structure

Category	WBS	Basis	Formula
Personnel	WP-1.1 to WP-12.5	PLANNING PLACEHOLDER - NOT APPROVED	Person-weeks from Resource_Plan x unit cost
Infrastructure	WP-10.2	PLANNING PLACEHOLDER - NOT APPROVED	Quantity x unit cost
Software	WP-6.9	PLANNING PLACEHOLDER - NOT APPROVED	Licences x unit cost
Data storage	WP-5.2	PLANNING PLACEHOLDER - NOT APPROVED	Storage volume x unit cost
AI development	WP-7.3, WP-8.3	PLANNING PLACEHOLDER - NOT APPROVED	Compute and API usage x unit cost
Testing and security	WP-9.3	PLANNING PLACEHOLDER - NOT APPROVED	Test effort x unit cost
Deployment	WP-10.3	PLANNING PLACEHOLDER - NOT APPROVED	Deployment effort x unit cost
Training	WP-11.2 to WP-11.5	PLANNING PLACEHOLDER - NOT APPROVED	Sessions x unit cost

Contingency	All	AS-08	10% of direct cost (AS-08)
Management reserve	All	AS-08	5% of direct cost (AS-08)

Direct total = sum of cost lines C-01 to C-22. Contingency (C-23) = 0.10 × direct total. Reserve (C-24) = 0.05 × direct total. Grand total (C-25) = direct + contingency + reserve. Approved amount and actual spend stay blank until evidence exists.

7.4 Resource Conflicts

Potential conflicts: the 0.5 FTE Clinical Lead is needed across P6, P7, P8 and P9 in overlapping weeks; Data Engineers are shared between integration (P5) and testing (P9); and P6, P7 and P8 overlap in Weeks 13–16. Mitigation: prioritise Must stories, stagger clinical reviews in the weekly slot (AS-15) and escalate unfilled roles under R-12.

8 Risk Management

8.1 Approach

Each risk is scored Probability (1–5) × Impact (1–5) = 1–25. Ratings: Low 1–4, Medium 5–9, High 10–14, Critical 15–25. The register has 25 risks, each with owner, mitigation, contingency, early-warning indicator and trigger (workbook sheet Risk_Register).

8.2 Top 15 Risks

R-ID	Event	Category	P	I	Score	Rating	Owner
R-01	Source data quality is poor	Data	4	5	20	Critical	Data Lead (TBC)

R-02	Scope creep	Scope	3	3	9	Medium	Project Manager
R-03	Budget overrun	Cost	3	4	12	High	Finance Lead (TBC)
R-04	Delayed stakeholder approvals	Governance	3	4	12	High	Executive Sponsor (TBC)
R-05	Regulatory or data-protection non-compliance	Compliance	3	5	15	Critical	Legal/DPO
R-06	Cybersecurity threat	Security	3	5	15	Critical	IT/Security Lead
R-07	Clinical user change resistance	People	4	3	12	High	Change Lead
R-08	AI model inaccuracy	AI	3	4	12	High	ML Lead (TBC)
R-09	Vendor delay	Procurement	3	3	9	Medium	Project Manager
R-10	Insurance data unavailability	Data	3	4	12	High	Finance Lead (TBC)
R-11	Privacy breach or PHI exposure	Security	2	5	10	High	Legal/DPO

R-12	Resource shortage	Resource	3	3	9	Medium	Project Manager
R-13	Schedule delays	Schedule	4	3	12	High	Project Manager
R-14	Integration complexity	Technical	4	4	16	Critical	Data Lead (TBC)
R-15	Low user adoption	People	3	4	12	High	Change Lead

8.3 Additional and Cross-Track Risks

R-16 to R-20 cover infrastructure or cloud failure, data residency non-compliance, unsafe or hallucinated AI output, executive disengagement and knowledge loss at handover. R-21 to R-25 are cross-track risks arising because inputs from other tracks are provisional.

R-ID	Event	P	I	Score	Rating	Owner
R-21	ML model performance insufficient	3	4	12	High	ML Lead (TBC)
R-22	Missing label data	3	4	12	High	Data Lead (TBC)
R-23	Unreliable table relationships discovered late	3	4	12	High	Data Lead (TBC)
R-24	Discrepancy across cross-track repositories	3	3	9	Medium	Project Manager

R-25	Participant readiness claimed but unverifiable	3	3	9	Medium	Project Manager

8.4 Probability–Impact Matrix

Risk IDs are placed at their (Probability, Impact) position. Cells show the register score band by position: score 15–25 Critical, 10–14 High, 5–9 Medium, 1–4 Low. Empty cells show a dash.

P \ I	1	2	3	4	5
5	-	-	-	-	-
4	-	-	R-07, R-13	R-14	R-01
3	-	-	R-02, R-09, R-12, R-20, R-24, R-25	R-03, R-04, R-08, R-10, R-15, R-21, R-22, R-23	R-05, R-06
2	-	-	-	R-16, R-19	R-11, R-17, R-18
1	-	-	-	-	-

Rating thresholds: Low 1–4 (light green) | Medium 5–9 (yellow) | High 10–14 (orange) | Critical 15–25 (red). Risk IDs are positioned by their Probability and Impact values from Risk_Register (R-01 to R-25).

8.5 Top 5 Priority Risks

R-01 - Source data quality is poor (score 20, Critical; owner Data Lead (TBC)). Mitigation: Profile all 27 datasets early; agree cleaning rules; quality gate at MS-03. Trigger: DQ coverage below target at MS-03. Early warning: Profiling error rate above threshold.

R-14 - Integration complexity (score 16, Critical; owner Data Lead (TBC)). Mitigation: Phased integration; read-only; early FK map. Trigger: Pipeline failures at W9. Early warning: Failed extractions.

R-05 - Regulatory or data-protection non-compliance (score 15, Critical; owner Legal/DPO). Mitigation: Early legal review; DPIA before deployment. Trigger: DPIA not started by W10. Early warning: Open legal findings.

R-06 - Cybersecurity threat (score 15, Critical; owner IT/Security Lead). Mitigation: RBAC, MFA, encryption, pen test. Trigger: Critical finding open. Early warning: Security scan alerts.

R-07 - Clinical user change resistance (score 12, High; owner Change Lead). Mitigation: Champions; co-design; training. Trigger: Training completion below target. Early warning: Low attendance.

9 Communication and Quality Management

9.1 Communication Plan

Audience	Information	Channel	Frequency	Owner
Executive Board	Executive status report	Report and briefing	Monthly	Project Manager
Steering Committee	Progress, risks, decisions, gate readiness	SteerCo meeting	Bi-weekly	Project Manager
PMO	Schedule, budget, risk, issues	PMO meeting	Weekly	Project Manager

Sprint teams	Sprint goal, backlog, impediments	Planning, daily stand-up, review, retrospective	Per sprint (2 weeks)	Business Analyst
Clinical users	Design reviews, safety checks	Clinical review session	Weekly	Clinical Lead (TBC)
Ministry of Health	Compliance and progress summary	Briefing note	Quarterly	Legal/DPO
All staff	Project news, training dates	Email and notice board	Monthly	Change Lead
Risk owners	Risk status and actions	Risk update	Weekly	Project Manager
Trainees	Training schedule, materials	Training sessions	Per session	Change Lead
Project team	Status, actions, blockers	Weekly project meeting	Weekly	Project Manager

9.2 Communication Templates

· Weekly status report (RAG, progress, risks, next steps)

· Meeting agenda

· Minutes

· Action log

9.3 Escalation Process

Level	From	To	Trigger
L1	Workstream lead	Project Manager	Issue unresolved within 2 working days
L2	Project Manager	PMO	Variance Amber or risk High
L3	PMO	Steering Committee	Variance Red, Critical risk or gate threat
L4	Steering Committee	Executive Board / Sponsor	Decision beyond SteerCo authority

9.4 Quality Management Plan

Area	Objective	Metric	Target	Evidence
Data quality	Trustworthy integrated data	Dataset profiling coverage; FK failure rate	100% of 27 datasets profiled; FK errors resolved	DQ report
Dashboard quality	Correct, usable dashboards	KPI reconciliation; UAT pass %	95% UAT pass (PROPOSED)	UAT sign-off
ML model quality	Models fit for purpose	Agreed metrics vs thresholds	Meet thresholds agreed at W11 (TBC)	Evaluation and SHAP/LIME report

GenAI quality	Grounded, safe answers	Citation accuracy; hallucination rate	Targets agreed in Discovery (TBC)	AI evaluation report
Documentation	Complete, consistent documents	Review defects per document	Zero open critical defects	Reviewed documents
Testing	Defects found before release	Test pass %; open defects	No critical defects at Go/No-Go	Test summary
UAT	Users accept the solution	UAT pass %; sign-offs	95% pass (PROPOSED)	UAT sign-off
Security and privacy	No critical vulnerabilities; DPIA done	Findings by severity; DPIA status	No critical open; DPIA signed	Pen test report; DPIA
Accessibility	Usable by all staff	Contrast and keyboard checks	All checks passed	Checklist
Release readiness	Safe go-live	Readiness checklist completion	100% mandatory items	Go/No-Go pack

10 Agile Backlog and Sprint Plan

10.1 Product Backlog (35 user stories - full detail in workbook sheet Product_Backlog)

The Product Backlog contains 35 user stories across 10 epics. Priority distribution: Must = 18 | Should = 10 | Could = 5 | Won't = 2 (deferred to v2).

Each user story follows the format: As a [role], I want [capability], so that [business value]. Each carries US-ID, Epic link, User story narrative, Acceptance criteria in Given/When/Then format, MoSCoW priority, Story point estimate, Dependencies (including DEP-xx cross-track items where relevant), Proposed owner role, Sprint allocation (S1–S4), Status, Definition of Done, Related WBS work package ID.

Epic distribution:

· EP-1 Data Platform: US-001 to US-006, US-009, US-010, US-035 (Won't)

· EP-2 Executive Analytics: US-011, US-012, US-015, US-016, US-018, US-027

· EP-3 Clinical Analytics: US-020

· EP-4 Financial Analytics: US-013, US-014

· EP-5 Operations Analytics: US-021, US-022, US-024

· EP-6 Patient Experience: US-019, US-023

· EP-7 Predictive Models: US-025, US-026, US-029, US-030

· EP-8 GenAI Assistants: US-031, US-032, US-034 (Won't)

· EP-9 Governance and Security: US-007, US-008, US-017, US-028

· EP-10 Change and Training: US-033

Sample stories (first five shown here; the full 35 are in workbook sheet Product_Backlog):

US-001 | EP-1 | As a Data Engineer, I want to ingest extracts from the seven source systems read-only, so that the warehouse holds a complete raw copy. | Must | 8 pts | Sprint 1 | WP-5.2

US-002 | EP-1 | As a Data Lead, I want to profile all 27 core datasets, so that data quality is measured before integration. | Must | 5 pts | Sprint 1 | WP-3.3

US-003 | EP-1 | As a Data Engineer, I want to standardise categories and resolve duplicate records, so that KPIs use consistent values. | Must | 8 pts | Sprint 1 | WP-4.2

US-004 | EP-1 | As a Data Lead, I want to verify foreign key integrity and issue a data quality report, so that relationships can be trusted. | Must | 5 pts | Sprint 1 | WP-4.4

US-005 | EP-1 | As a Data Engineer, I want to build the integrated warehouse model, so that analysts query one source of truth. | Must | 13 pts | Sprint 1 | WP-5.2

US-034 (SMS discharge notifications) and US-035 (real-time streaming) are marked Won't for v1 and explicitly deferred to v2 via CHG-01 and SC-OUT-10 respectively.

Note: US-007 and US-008 sit in EP-9; US-018 (monthly KPI narrative pack) is assigned to EP-2 so that all 35 stories belong to one epic.

10.2 Sprint Plan

Four two-week sprints run in W9–W16 (AS-03). Capacity: 40 points per sprint (PROPOSED - velocity to be confirmed after Sprint 1). Workbook sheet Product_Backlog computes planned points per sprint.

SPR-01 - W9–W10

Item	Detail
Sprint goal	Integrated data foundation ready for analytics
Dates (weeks)	W9–W10 (Timeline B)
Stories	US-001 to US-008
Capacity assumption	40 points per sprint (PROPOSED - velocity to be confirmed after Sprint 1)
Dependencies	Source extracts available; FK map (DEP-02); access granted (AS-04)
Deliverables	Integrated warehouse; DQ and reconciliation reports
Responsibilities	Data Engineer, Data Lead, IT/Security Lead
Definition of Done	Acceptance criteria met and verified by named reviewer; code/config reviewed; documentation updated; tests passing; security/privacy reviewed where applicable; deployed to staging.

Planning activities	Sprint planning on day 1: confirm goal, select stories, check capacity
Daily coordination	Daily 15-minute coordination
Review	Sprint Review with Product Owner role and stakeholders on last day
Retrospective	Sprint Retrospective after the Review
Risks / blockers	Planned load 50 points (10 over the 40-point assumption, PROPOSED); US-009 and US-010 (Should) moved to Sprint 2 and further carry-over is expected if velocity is below plan; Source access delays; FK errors
Exit criteria	Data ready for dashboards; FK integrity passed; integration specification signed

SPR-02 - W11–W12

Item	Detail
Sprint goal	First two dashboards live (Executive and Financial)
Dates (weeks)	W11–W12 (Timeline B)
Stories	US-009 to US-015, US-017 to US-019
Capacity assumption	40 points per sprint (PROPOSED - velocity to be confirmed after Sprint 1)

Dependencies	Warehouse (SPR-01); KPI definitions (DEP-08)
Deliverables	Executive and Financial dashboards
Responsibilities	Dashboard Developer, Business Analyst
Definition of Done	Acceptance criteria met and verified by named reviewer; code/config reviewed; documentation updated; tests passing; security/privacy reviewed where applicable; deployed to staging.
Planning activities	Sprint planning on day 1: confirm goal, select stories, check capacity
Daily coordination	Daily 15-minute coordination
Review	Sprint Review with Product Owner role and stakeholders on last day
Retrospective	Sprint Retrospective after the Review
Risks / blockers	Planned load 39 points; KPI definitions unsigned; claims data (R-10)
Exit criteria	Sprint Review with Sponsor; demo sign-off

SPR-03 - W13–W14

Item	Detail

Sprint goal	Clinical, Operations and Patient Experience dashboards; first ML model evaluated
Dates (weeks)	W13–W14 (Timeline B)
Stories	US-016, US-020 to US-028
Capacity assumption	40 points per sprint (PROPOSED - velocity to be confirmed after Sprint 1)
Dependencies	Labels (DEP-04); features (DEP-06)
Deliverables	Three dashboards; model 1 evaluation
Responsibilities	Dashboard Developer, ML Engineer, Legal/DPO
Definition of Done	Acceptance criteria met and verified by named reviewer; code/config reviewed; documentation updated; tests passing; security/privacy reviewed where applicable; deployed to staging.
Planning activities	Sprint planning on day 1: confirm goal, select stories, check capacity
Daily coordination	Daily 15-minute coordination
Review	Sprint Review with Product Owner role and stakeholders on last day
Retrospective	Sprint Retrospective after the Review

Risks / blockers	Planned load 40 points; Labels missing (R-22)
Exit criteria	Dashboards UAT-signed; model 1 evaluated

SPR-04 - W15–W16

Item	Detail
Sprint goal	All 3 ML models evaluated and GenAI pilots ready
Dates (weeks)	W15–W16 (Timeline B)
Stories	US-029 to US-033
Capacity assumption	40 points per sprint (PROPOSED - velocity to be confirmed after Sprint 1)
Dependencies	Model evaluation (DEP-05); prompt library (DEP-10)
Deliverables	Models 2 and 3; explainability; prompt library; RAG pilot
Responsibilities	ML Engineer, GenAI Engineer, Change Lead
Definition of Done	Acceptance criteria met and verified by named reviewer; code/config reviewed; documentation updated; tests passing; security/privacy reviewed where applicable; deployed to staging.

Planning activities	Sprint planning on day 1: confirm goal, select stories, check capacity
Daily coordination	Daily 15-minute coordination
Review	Sprint Review with Product Owner role and stakeholders on last day
Retrospective	Sprint Retrospective after the Review
Risks / blockers	Planned load 39 points; Model performance (R-21); unsafe output (R-18)
Exit criteria	Model Review Gate ready; GenAI ready for evaluation

10.3 Relationship Between Sprints and Broader Schedule

Sprints cover the build work in W9–W16. Data collection and cleaning (W5–W8) run as waterfall. Testing and deployment are hybrid: system integration, security testing and UAT are not sprint-native and follow the test plan, while fixes are taken through the backlog.

11 Change Management

11.1 Change Control Process

1. Raise a change request (CR form) with description and reason.

2. PMO logs it in Change_Log and assesses impact on scope, cost, schedule, quality, security and resources.

3. Classify as Minor, Standard or Major.

4. Approve, reject or defer according to authority.

5. Update baselines, communicate and track to closure.

Class	Criteria	Authority (TBC)
Minor	No effect on scope, cost or milestones	Project Manager
Standard	Affects a work package or sprint backlog, within contingency	PMO / Sponsor role
Major	Affects scope baseline, a Stage Gate, budget beyond contingency or security	Steering Committee

11.2 Illustrative Change Request CHG-01

ILLUSTRATIVE - not a real request.

Field	Detail
ID	CHG-01
Title	Add SMS discharge notifications (ILLUSTRATIVE)
Description	Illustrative request to add SMS discharge notifications to the platform
Classification	Major
Impact	Scope: adds US-034; Cost: new messaging cost; Schedule: not in W9–W16 capacity; Quality: new tests; Security: patient contact data; Resource: GenAI Engineer

Decision	Deferred to v2 (ILLUSTRATIVE decision)
Authority	SteerCo (TBC)
Status	ILLUSTRATIVE

11.3 Adoption Strategy

· Clinical and executive champions in each facility

· Clinical co-design of dashboards and assistants

· Role-based training in English, French and Kinyarwanda

· Feedback channels and sprint review input

· Adoption metrics (active users, training completion)

11.4 Resistance Management

Source	Mitigation
Clinicians worried about workflow and AI trust	Co-design, explainability and human oversight (AS-09)
Administrators worried about extra reporting work	Show time saved; simplify inputs
IT staff worried about security and support load	Joint design reviews; runbooks; hypercare

Executives focused on cost	Benefit tracking (BEN-01 to BEN-09) and regular value updates

12 Monitoring and Control

12.1 PMO Dashboard Specification

Category	Indicators
Schedule performance	SPI, milestone on-time %, critical path slip
Budget performance	CPI, budget variance %, forecast at completion
Risk status	Critical risks open, top-5 trend
Issue log	Open issues by severity and age
Milestone progress	MS-01 to MS-11 status
Resource utilisation	FTE planned vs available
Quality metrics	UAT pass %, data quality coverage %
Stakeholder satisfaction	Pulse survey score

12.2 KPI Definitions

KPI	Formula	Target	Green	Amber	Red
KPI-001 SPI	EV / PV	1.00 or above	0.95 or above	0.85 to 0.94	Below 0.85
KPI-002 CPI	EV / AC	1.00 or above	0.95 or above	0.90 to 0.94	Below 0.90
KPI-003 Milestone on-time %	Milestones on time / milestones due	90% or above	90% or above	75% to 89%	Below 75%
KPI-004 Critical risks open	COUNTIF(Rating=Critical, status open)	Fewer than 2	0 to 1	2 to 3	4 or more
KPI-005 UAT pass %	Passed / executed	95% or above	95% or above	85% to 94%	Below 85%
KPI-006 Budget variance %	(FAC − Approved) / Approved	Within 5%	Within 5%	5% to 10%	Above 10%
KPI-007 Training completion %	Completed / enrolled	90% or above	90% or above	75% to 89%	Below 75%
KPI-008 Data quality coverage %	Datasets passing / 27	100%	95% or above	85% to 94%	Below 85%

12.3 Reporting Cadence

Weekly: PMO status and risk update. Bi-weekly: Steering Committee. Monthly: Executive Board report and budget review. Quarterly: Ministry of Health briefing. The PMO dashboard in workbook sheet Project_Dashboard is ILLUSTRATIVE only.

12.4 Tool Recommendation

Recommended: the Control Workbook for planning data, Power BI for the PMO dashboard (consistent with the assignment's dashboard tooling), and Jira or Trello for sprint boards. Microsoft Project may be used if the organisation already licenses it. The choice is TBC.

13 Responsible AI, Privacy, Security and Governance

13.1 Principles

1. Human oversight
2. Data minimisation
3. Transparency
4. Fairness
5. Privacy
6. Safety
7. Traceability
8. Governance

13.2 AI Governance Structure

Body	Remit
AI Governance Committee	Approves AI use cases, reviews evaluation and RAI pack

Clinical Review Committee	Reviews clinical content, safety and oversight design
Data Protection Office	Owns DPIA, privacy and regulatory review

13.3 Privacy and Data Protection

Collect and expose the minimum data; pseudonymise identifiers in analytics and AI layers (REQ-DR-006); host data in Rwanda (AS-10); complete a DPIA before deployment (REQ-RAI-009). No patient data is included in this package.

13.4 Security

Role-based access, multi-factor authentication, encryption at rest and in transit, audit logging, secrets management, vendor security review, incident response and penetration testing before Go/No-Go (REQ-SEC-001 to REQ-SEC-008).

13.5 Regulatory Review

Legal/DPO would confirm the applicable Rwandan data-protection requirements (including the national personal data protection law) and health-sector rules during Discovery. Applicability is TBC; this report gives no legal advice.

13.6 Model Monitoring and Retraining

Each model has documented metrics, drift monitoring and retraining triggers (REQ-RAI-005, REQ-RAI-008). GenAI assistants are monitored for citation accuracy and unsafe output (REQ-RAI-004).

13.7 Clinical Oversight

AI outputs are decision support only. A clinician reviews any output affecting care, and a finance reviewer reviews outputs affecting financial decisions (AS-09, REQ-RAI-001, REQ-RAI-007).

14 Project Closure Approach

14.1 Closure Criteria

ID	Criterion	Evidence source
CLO-01	All 19 deliverables verified against acceptance criteria	Deliverables_Tracker
CLO-02	All requirements REQ traced to passed AC	Requirements_Traceability
CLO-03	Stage Gate 3 Go/No-Go approved and deployment completed	Decision_Log
CLO-04	Budget reconciled; actual spend vs approved reported	Budget
CLO-05	Schedule performance reported against baseline	Project_Schedule
CLO-06	Open issues and risks transferred or closed	Issue_Log; Risk_Register
CLO-07	Handover to operations signed	Handover pack
CLO-08	Benefits tracking (BEN-01 to BEN-09) set up for 12 months	KPI_Definitions
CLO-09	Lessons learned (LL-01 to LL-15) captured	Lessons_Learned
CLO-10	Formal sign-off by Sponsor role (TBC)	Closure report

14.2 Handover

· Operational runbooks and support model

· Data dictionary, lineage and pipeline documentation

· Model documentation, monitoring and retraining procedures

· Security and access administration

· Training materials and champion network

14.3 Benefits Realisation

Benefits BEN-01 to BEN-09 would be tracked for 12 months after closure, using the KPIs in Section 2.4 and Section 12.2. The closure report would state benefits as realised only where evidence exists.

14.4 Lessons Learned

Categories: Planning, Requirements, Scope, Risk, Cross-track, Schedule, Budget, Agile, Quality, Governance, Communication, Ethics, Documentation. LL-01 to LL-15 from the planning exercise (Timeline A) are in workbook sheet Lessons_Learned; they are not implementation lessons.

14.5 Closure Report Structure

· Project summary

· Deliverables completed

· Budget performance

· Schedule performance

· Lessons learned

· Outstanding issues

· Recommendations for future projects

· Formal sign-off

The closure report would be a template until Timeline B reaches Week 24. No closure data exists today.

15 Conclusions and Recommendations

15.1 Conclusions

1. The assignment can be met by a phased, governed 24-week programme (Option B).
2. Data quality and integration are the main delivery risks.
3. Responsible AI controls must be gate conditions, not afterthoughts.
4. Cross-track inputs are provisional and must be evidenced.
5. All figures in this package are proposals and placeholders.

15.2 Recommendations

1. Approve Option B.
2. Confirm the 24-week duration.
3. Confirm read-only integration for v1.
4. Appoint the Sponsor, Clinical Lead and Data Lead before Gate 1.
5. Have Finance validate the placeholder budget.
6. Start Discovery with profiling of the 27 datasets.
7. Attach evidence for each DEP item before its need-by week.

15.3 What Is Not Claimed

- No implementation, deployment or approval
- No actual expenditure or measured savings
- No model accuracy figures
- No completed milestones
- No named stakeholders or contracts
- No benefit realisation

16 References and Source Notes

Ref	Source	Use
S1	AnalystLab Africa Week 6 Official Assignment PDF (pp. 1–40)	Primary source of truth
S2	PMBOK Guide concepts	Methodological alignment (not certified)
S3	Agile Practice Guide concepts	Sprint plan methodology
S4	Scrum Guide concepts	Sprint ceremony terminology
S5	Participant reference repositories (Data Science and ML Engineering, GitHub) - links recorded for audit-trail completeness; not reviewed at README level within the Week 6 submission window	Recorded as candidate provisional cross-track inputs; NOT treated as project facts
S6	Participant LinkedIn progress posts - links recorded for audit-trail completeness; not reviewed within the Week 6 submission window	Recorded for completeness; NOT treated as project facts
S7	Responsible AI principles (generic)	Design of RAI requirements

16.1 Integrity Statement

No fabricated data is included. Official assignment facts, proposed planning assumptions (AS-xx), illustrative examples (labelled ILLUSTRATIVE) and confirmed actuals (blank until evidence exists) are kept distinct. Cross-track inputs from the Data Analytics, Data Science, ML and GenAI tracks are PROVISIONAL; every DEP-xx has the status Awaiting input until an evidence item is attached.

16.2 Recommended Next Action

The first post-submission action is to review S5 and S6 and attach evidence items to the Cross-Track Dependency Register.

17 Appendix A - Traceability Summary

BO	Requirements (REQ- prefix omitted)	Work packages and activities	Deliverables	AC
BO-01	F-001, F-002, F-003, F-004, F-005, NF-003, NF-009, NF-010, DR-001, DR-002, DR-003, DR-004, DR-005, DR-007, DR-008, SEC-003, SEC-006	WP-2.4, WP-3.3 (ACT-3.3.1), WP-3.4 (ACT-3.4.1), WP-3.5, WP-4.1 (ACT-4.1.1), WP-4.2 (ACT-4.2.1), WP-4.4 (ACT-4.4.1), WP-5.1 (ACT-5.1.1), WP-5.3 (ACT-5.3.1), WP-5.4 (ACT-5.4.1), WP-5.5, WP-9.4, WP-10.2	D-05, D-09, D-12	AC-01, AC-02, AC-03, AC-04, AC-08, AC-09
BO-02	F-006, NF-001	WP-6.3 (ACT-6.3.1), WP-9.4	D-12, D-13	AC-05
BO-03	F-008, F-010, DR-006, RAI-001, RAI-007, RAI-009	WP-5.3 (ACT-5.3.1), WP-6.5 (ACT-6.5.1), WP-6.7, WP-8.7 (ACT-8.7.1)	D-09, D-12, D-13	AC-05, AC-08
BO-04	F-007	WP-6.4 (ACT-6.4.1)	D-13	AC-05
BO-05	F-009, F-020, NF-002, NF-006, NF-008, OP-001, OP-002, OP-003, OP-004	WP-6.6, WP-10.1 (ACT-10.1.1), WP-10.2, WP-10.4 (ACT-10.4.1), WP-11.2, WP-12.2	D-12, D-13, D-15, D-17	AC-05, AC-10, AC-11, AC-12

BO-06	F-011, F-018, F-019, NF-004, NF-005, NF-007, SEC-001, SEC-002, SEC-004, SEC-005, SEC-007, SEC-008, RAI-010	WP-1.5 (ACT-1.5.1), WP-2.14 (ACT-2.14.1), WP-6.8 (ACT-6.8.1), WP-6.9, WP-8.4, WP-9.3 (ACT-9.3.1), WP-9.5 (ACT-9.5.1), WP-10.1 (ACT-10.1.1), WP-10.2, WP-12.1 (ACT-12.1.1)	D-09, D-12, D-13, D-16	AC-05, AC-07, AC-08, AC-09, AC-11, AC-12
BO-07	F-015, F-016, F-017, RAI-004, RAI-006	WP-8.1 (ACT-8.1.1), WP-8.3 (ACT-8.3.1), WP-8.4, WP-8.5, WP-8.6 (ACT-8.6.1)	D-09, D-13	AC-07
BO-08	F-012, F-013, F-014, RAI-002, RAI-003, RAI-005, RAI-008	WP-7.3 (ACT-7.3.1), WP-7.4 (ACT-7.4.1), WP-7.5 (ACT-7.5.1), WP-7.6, WP-7.7	D-09, D-13	AC-06, AC-08

18 Appendix B - Cross-Track Dependency Register

The Cross-Track Dependency Register records 18 inputs expected from the Data Analytics, Data Science, ML Engineering and Generative AI tracks. All 18 dependencies have the status "Awaiting input" - none has an evidence item attached. Full detail is available in workbook sheet Cross_Track_Dependency_Register.

DEP-ID	Dependency	Providing Track	Receiving Track	Need-by Week	Status
DEP-01	Data Quality Report	Data Science	Project Management	Week 8	Awaiting input

DEP-02	Relationship and FK documentation	Data Analytics	Project Management	Week 6	Awaiting input
DEP-03	Cleaned integrated dataset	Data Analytics	Data Science	Week 8	Awaiting input
DEP-04	Label availability	Data Science	ML Engineering	Week 11	Awaiting input
DEP-05	Model evaluation	ML Engineering	Project Management	Week 15	Awaiting input
DEP-06	Feature engineering specification	Data Science	ML Engineering	Week 12	Awaiting input
DEP-07	Pilot executive dashboard	Data Analytics	Project Management	Week 12	Awaiting input
DEP-08	KPI definitions	Data Analytics	Generative AI	Week 10	Awaiting input
DEP-09	Explainability report (SHAP/LIME)	ML Engineering	Project Management	Week 15	Awaiting input
DEP-10	Prompt library (20+ prompts)	Generative AI	Project Management	Week 14	Awaiting input
DEP-11	RAG architecture diagram	Generative AI	Project Management	Week 14	Awaiting input

DEP-12	Model API documentation	ML Engineering	Generative AI	Week 15	Awaiting input
DEP-13	AI evaluation report	Generative AI	Project Management	Week 17	Awaiting input
DEP-14	Responsible AI assessment	ML Engineering + Generative AI	Project Management	Week 17	Awaiting input
DEP-15	Business insights and KPI results	Data Analytics	Project Management	Week 13	Awaiting input
DEP-16	Forecast and statistical analysis	Data Science	Project Management	Week 14	Awaiting input
DEP-17	Deployment architecture	ML Engineering	Project Management	Week 16	Awaiting input
DEP-18	Final track submissions	All technical tracks	Project Management	Week 19	Awaiting input

Gate-blocking dependencies: DEP-01 (blocks MS-03), DEP-04 (blocks model scoping), DEP-05 (blocks MS-06), DEP-10 and DEP-13 (block MS-07), DEP-14 (blocks MS-08).

Integrity note: No cross-track input is treated as accepted until an evidence item is attached and the acceptance criterion is met. All DEP-01 to DEP-18 remain "Awaiting input" at the date of submission.

19 Appendix C - Risk Register Extract

R-ID	Event	Score	Rating	Owner
R-01	Source data quality is poor	20	Critical	Data Lead (TBC)
R-02	Scope creep	9	Medium	Project Manager
R-03	Budget overrun	12	High	Finance Lead (TBC)
R-04	Delayed stakeholder approvals	12	High	Executive Sponsor (TBC)
R-05	Regulatory or data-protection non-compliance	15	Critical	Legal/DPO
R-06	Cybersecurity threat	15	Critical	IT/Security Lead
R-07	Clinical user change resistance	12	High	Change Lead
R-08	AI model inaccuracy	12	High	ML Lead (TBC)
R-09	Vendor delay	9	Medium	Project Manager
R-10	Insurance data unavailability	12	High	Finance Lead (TBC)
R-11	Privacy breach or PHI exposure	10	High	Legal/DPO

R-12	Resource shortage	9	Medium	Project Manager
R-13	Schedule delays	12	High	Project Manager
R-14	Integration complexity	16	Critical	Data Lead (TBC)
R-15	Low user adoption	12	High	Change Lead
R-16	Infrastructure or cloud failure	8	Medium	IT/Security Lead
R-17	Data residency non-compliance	10	High	Legal/DPO
R-18	Unsafe or hallucinated AI output	10	High	GenAI Lead (TBC)
R-19	Executive disengagement	8	Medium	Project Manager
R-20	Knowledge loss at handover	9	Medium	Project Manager
R-21	ML model performance insufficient	12	High	ML Lead (TBC)
R-22	Missing label data	12	High	Data Lead (TBC)
R-23	Unreliable table relationships discovered late	12	High	Data Lead (TBC)
R-24	Discrepancy across cross-track repositories	9	Medium	Project Manager

R-25	Participant readiness claimed but unverifiable	9	Medium	Project Manager

The full risk register with mitigation, contingency, trigger and early-warning indicators is in workbook sheet Risk_Register. The 25 risks R-01 to R-25 are extracted above. The top 5 priority risks (R-01, R-14, R-05, R-06, R-07) are discussed in Section 8.5.

20 Appendix D - Requirements Compliance Matrix

Assignment requirement	Where addressed	Status
Task 1 Business Case	Section 2; Deliverable D-01	Compliant
Task 2 Project Charter	Section 3; D-02	Compliant
Task 3 Stakeholder Register	Section 4; D-03	Compliant
Task 4 Scope Management	Section 5; D-04	Compliant
Task 5 Work Breakdown Structure	Section 6.1; Workbook WBS; D-05	Compliant
Task 6 Project Schedule and Gantt	Section 6.2–6.5; Workbook Project_Schedule, Gantt_Chart; D-06	Compliant
Task 7 Resource Plan	Section 7.1–7.2; Workbook Resource_Plan; D-07	Compliant

Task 8 Budget Planning	Section 7.3; Workbook Budget; D-08	Compliant
Task 9 Risk Register (15+ risks)	Section 8.2–8.3; Workbook Risk_Register; D-09	Compliant
Task 10 Probability–Impact Matrix and top five	Section 8.4–8.5; Workbook Risk_Matrix; D-10	Compliant
Task 11 Communication Plan	Section 9.1–9.3; D-11	Compliant
Task 12 Quality Management Plan	Section 9.4; D-12	Compliant
Task 13 Product Backlog (25+ stories, MoSCoW)	Section 10.1; Workbook Product_Backlog; D-13	Compliant
Task 14 Sprint Planning (four two-week sprints)	Section 10.2; Workbook Sprint_Plan; D-14	Compliant
Task 15 Change Management Plan	Section 11; Workbook Change_Log; D-15	Compliant
Task 16 Project Dashboard	Section 12; Workbook Project_Dashboard; D-16	Compliant
Task 17 Project Closure Report	Section 14; D-17	Compliant
Ethical, legal and regulatory considerations	Section 13	Compliant

Executive Summary (3–5 pages)	Section 1; D-18	Compliant
Presentation Slides (10–15 slides)	Executive Presentation (14 slides); D-19	Compliant
Project objectives coverage	Section 3.3; Appendix A	Compliant
Assessment criteria alignment	Sections 2–14 map to the six PM assessment areas	Compliant